

SIMATS ENGINEERING

CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 10%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Code of Conduct:

I Dillilokesh D 192311097(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dillilokesh D

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

ANSWERS

Task 1: Article Summary

(a) Full Citation, Domain and Problem Addressed

Albert, A. J., Murugan, R., & Sripriya, T. (2023). Diagnosis of heart disease using oversampling methods and decision tree classifier in cardiology. *Research on Biomedical Engineering*, 39(1), 99–113.

<https://doi.org/10.1007/s42600-022-00253-9>

Domain / Application Area: Biomedical engineering and clinical decision support, specifically automated diagnosis of cardiovascular (heart) disease using patient health records.

Specific ML Problem: This is a binary classification problem — predicting whether a patient has heart disease (presence/absence) from clinical attributes such as age, cholesterol, blood pressure, chest pain type, and ECG results. The paper treats it as a classic supervised learning task, but the real challenge it tackles is class imbalance: in real clinical datasets, the number of "disease" cases is usually much smaller than "no disease" cases, and a plain decision tree tends to get biased toward the majority class.

(b) Summary of Key Objective (my own words)

I picked this paper because it does not just apply a decision tree and stop there — it tries to fix a very practical problem that most textbook examples ignore, which is imbalanced medical data. The authors noticed that heart-disease datasets like the Framingham Heart Study (FHS) dataset are heavily skewed: the number of healthy patients far outnumbers the number of patients who actually have heart disease. When a CART decision tree is trained directly on such data, it becomes very good at predicting "no disease" simply because that class dominates, but it performs poorly at catching the actual disease cases, which is exactly the outcome that matters most in a hospital setting.

The research gap they identified is that most earlier heart-disease prediction studies reported high overall accuracy but did not pay enough attention to recall and f1-score, which are the metrics that really tell you whether the model is missing sick patients. To fill this gap, the authors proposed a combined pipeline they call the Balanced and Optimized Machine Learning (BOML) algorithm. It brings together oversampling (SMOTE and similar techniques) to balance the classes, attribute pruning to drop weak or redundant features, a CART decision tree as the core classifier, and rule pruning through hyper-parameter tuning to avoid overfitting. The objective was to show that this balanced pipeline diagnoses heart disease more reliably than a plain decision tree trained on unbalanced data, especially on the metrics that matter clinically.

Task 2: Methodology Analysis

(a) ML Technique(s) Used and Dataset Description

Technique: The core classifier is CART (Classification and Regression Tree), a decision-tree variant that splits nodes using the Gini impurity index and can handle both categorical and continuous attributes. Around this core classifier, the authors wrap several pre- and post-processing steps to form the BOML pipeline:

- Oversampling of the minority (disease-positive) class, mainly using SMOTE (Synthetic Minority Oversampling Technique), which generates synthetic samples rather than simply duplicating existing minority-class rows.
- Attribute pruning, where less informative or redundant clinical features are removed before training so the tree does not split on noise.
- CART decision tree induction on the balanced, pruned data.
- Rule pruning through hyper-parameter tuning (controlling tree depth, minimum samples per leaf, and similar parameters) to reduce overfitting of the tree to the training data.

Dataset: Two publicly available clinical datasets were used. The first is the Framingham Heart Study (FHS) dataset, which contains around 4,240 patient records with 15 attributes (age, sex, smoking status, blood pressure, cholesterol, glucose, BMI, and outcome label, among others) and is highly imbalanced. The second is the Statlog Heart dataset, a smaller and comparatively well-known benchmark with 270 patient records and 13 attributes, which is only lightly imbalanced. Using both a large/highly-imbalanced dataset and a small/lightly-imbalanced one was a deliberate choice by the authors so they could check whether their BOML pipeline behaves consistently regardless of how skewed the data is. The data was split for training and testing and validated using 10-fold cross-validation, with the class ratio of majority to minority samples reported before and after oversampling to show how much the balancing step actually corrected the skew.

(b) Mapping to CO4 Topics

This paper maps most directly onto two of our CO4 topics at once. First, it is a clear example of Decision Trees, since CART is the classifier that actually makes the diagnosis, using recursive binary splitting on attributes to build an interpretable tree of if-then rules. Second, it is an example of Inductive Learning in the broader sense, because the whole pipeline is about generalising a decision rule from a finite set of labelled patient records to unseen patients, which is the basic definition of inductive learning we studied in class. There is no Reinforcement Learning component here, and no probabilistic model like Naïve Bayes or Logistic Regression is used as the primary classifier, although the class-imbalance problem the authors are solving is conceptually related to the kind of statistical/probability-based reasoning we cover under Statistical Learning Methods.

Methodological Strength:

The biggest strength I noticed is that the authors did not just report accuracy. They deliberately reported precision, recall, f1-score and AUC-ROC together, and even explained why recall matters more than raw accuracy in a medical diagnosis setting (missing a sick patient is far more costly than a false alarm). Testing the pipeline on two datasets with very different imbalance ratios (FHS vs. Statlog) also strengthens the claim that BOML is not just tuned to one lucky dataset.

Methodological Limitation:

The main limitation is that the paper still relies on a single decision tree (CART) as the final classifier rather than comparing against ensemble tree methods like Random Forest or Gradient Boosted Trees, which are known to be more robust to overfitting and often perform better on medical tabular data. A single tree, even after pruning, can still be unstable — small changes in the training data can produce a fairly different tree structure — and the paper does not report variance across cross-validation folds, only aggregate metrics, so we cannot tell how stable the results actually are.

Task 3: Results & Critical Evaluation

(a) Key Results

The authors report that, compared with a plain CART decision tree trained on the original (unsampled) dataset, the BOML pipeline using SMOTE-balanced data gives a substantial jump across all four metrics they tracked. Specifically, they report an improvement of about 11% in accuracy, 75% in precision, 62% in recall, and 71% in f1-score on the more heavily imbalanced dataset, along with improved AUC-ROC curves for both the FHS and Statlog datasets. The gain was larger on the FHS dataset (which is more imbalanced) than on the Statlog dataset (which is only mildly imbalanced), which supports their central claim that the benefit of oversampling and pruning scales with how skewed the original data is.

Setting	Dataset Used	Reported Behaviour	Approx. Improvement (SMOTE + BOML vs. unsampled CART)
CART decision tree alone (no sampling, no pruning tuning)	Framingham Heart Study (FHS) and Statlog Heart datasets	Used as the baseline; lower recall and f1-score because the minority (disease-positive) class is under-represented	Baseline (0%)

BOML algorithm (SMOTE oversampling + attribute pruning + CART + rule pruning/hyper-parameter tuning)	Same two datasets, class-balanced with SMOTE before training	Consistently outperforms the baseline on all four metrics; gain is larger on the highly imbalanced FHS dataset than on the lightly imbalanced Statlog dataset	Accuracy $\approx +11\%$, Precision $\approx +75\%$, Recall $\approx +62\%$, F1-score $\approx +71\%$ (as reported by the authors)
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(Note: the source article presents exact per-fold accuracy/precision/recall/f1/AUC-ROC values in Tables 3 and 4 of the paper, comparing the BOML pipeline against the plain CART baseline on both datasets. The table above summarises the direction and approximate scale of improvement as reported by the authors in their abstract and results discussion; readers should consult the original tables in the article for the precise decimal figures.)

(b) Critical Evaluation of the Results

I think the reported metrics are plausible but should be read carefully rather than taken at face value. A jump of 75% in precision and 62% in recall sounds very large, and part of this is simply the arithmetic effect of starting from a low, imbalance-damaged baseline — when a plain decision tree's recall on the minority class is very poor to begin with, even a modest real improvement can look like a huge percentage gain. So the headline numbers are technically correct but can create an inflated impression if read without the baseline context.

There are also a few potential sources of bias worth flagging. First, SMOTE creates synthetic minority-class samples by interpolating between existing minority points, so if the original minority samples are not fully representative of real-world disease presentations, the synthetic samples inherit that same narrow coverage — the model could be learning patterns that are somewhat artificial rather than clinically diverse. Second, both datasets (FHS and Statlog) are decades-old, Western-population studies, so the learned tree may not generalise well to other populations, age groups, or more recent diagnostic criteria. Third, the paper does not report whether the class balancing (SMOTE) was applied only to the training folds or across the whole dataset before splitting; if SMOTE-generated synthetic points leaked into the test set, the reported performance could be optimistically biased, which is a common pitfall in imbalanced-learning papers.

To strengthen the findings, I would want to see: (i) statistical significance testing or confidence intervals across the 10 folds rather than single-point metrics, (ii) a comparison against ensemble methods (Random Forest, XGBoost) and at least one statistical baseline (Logistic Regression or Naïve Bayes) run under the exact same balancing pipeline, and (iii) external validation on a third, independent heart-disease dataset to check that the BOML pipeline is not simply overfitted to the quirks of FHS and Statlog.

(c) Real-World Applicability

A lightweight, interpretable model like a pruned CART tree is actually well suited to real clinical deployment because doctors can literally read the decision path (for example: "if age > 55 and cholesterol > 240 and resting ECG abnormal, then high risk"), which builds trust in a way that a black-box deep learning model does not. This kind of model could realistically be embedded in a primary-care screening tool, a triage checklist app, or a clinical decision-

support widget inside an electronic health record system, especially in resource-limited settings where a full battery of expensive diagnostic tests is not feasible and a quick, explainable risk flag is genuinely useful.

That said, scaling this from a research paper to an actual hospital system raises real challenges. Data availability is a big one: hospitals would need consistent, well-labelled records at the same scale and quality as FHS, and many regions simply do not have that kind of longitudinal health data digitised yet. There are also ethical and regulatory issues — a model trained mostly on one population's data could perform worse (and thus be unfair) on other ethnic groups, age ranges, or comorbidity profiles, so any deployment would need local revalidation and probably regulatory clearance as a medical device. Compute requirements are not really a barrier here, since decision trees are cheap to train and run, but the bigger practical hurdle is integration into clinical workflow, model monitoring for data drift as patient populations change over time, and making sure clinicians treat the tree's output as a decision aid rather than a final diagnosis.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Connected to CO4 Concepts

1. Class imbalance directly affects the inductive bias of a decision tree. Reading this paper made it much clearer to me why the inductive learning concept of "generalising from examples" breaks down when the examples are not representative in proportion. A CART tree trained on skewed data essentially learns that predicting the majority class is a safe bet, which is a very concrete illustration of how the training distribution shapes what a model "induces" as its decision rule.

2. Pruning is not just about accuracy, it is about controlling overfitting versus underfitting. The rule-pruning and hyper-parameter tuning step in BOML connects directly to what we studied about decision-tree pruning (pre-pruning via depth/leaf-size limits, and post-pruning). Seeing it applied to a real medical dataset helped me understand that pruning is what keeps a tree from memorising noise in patient records while still keeping the tree readable.

3. Evaluation metric choice is itself a modelling decision, not just a reporting formality. This connects to the statistical learning topics on precision, recall, F1-score, and ROC-AUC that we covered in class. The paper is a good reminder that in domains like healthcare, optimising for accuracy alone (a very statistical-learning-flavoured metric) can hide a model that is clinically dangerous, and that recall-oriented evaluation is often the more honest choice when false negatives carry a high real-world cost.

(b) Proposed Extension of the Research

If I were extending this work, I would propose replacing the single CART tree in the BOML pipeline with an ensemble of pruned trees — specifically a Random Forest or a Gradient Boosted Trees model — while keeping the same SMOTE-based balancing and attribute-pruning steps, and then testing the whole pipeline on a more recent, more diverse dataset such as the UCI/Kaggle combined heart-disease dataset that merges Cleveland, Hungary, Switzerland, Long Beach and Statlog records (roughly 1,190 patients from multiple populations).

Feasibility: This is very achievable with existing tools — scikit-learn already provides Random Forest, Gradient Boosting, SMOTE (via imbalanced-learn), and cross-validation utilities, so the experiment would not need new infrastructure, just a re-run of the same evaluation protocol the original authors used, plus the ensemble models as additional comparison points.

Expected Impact: I would expect the ensemble version to improve stability across folds (lower variance in the reported metrics) and possibly push recall and AUC-ROC even higher than the single-tree BOML pipeline, while a multi-population dataset would let us actually test the generalisability concern I raised in Task 3(b) instead of just speculating about it. The trade-off would be a small loss in interpretability compared to a single CART tree, so I would also suggest reporting feature-importance scores from the ensemble alongside the original tree's rules, so clinicians still get some explanation of why a patient was flagged as high-risk, even if the underlying model is now a forest instead of a single tree.